

Applications of Monge Maps

- Image and color transfer
 - Map color histograms or pixel distributions from one style to another (e.g., photo to painting)
- Domain alignment & dataset matching
 - Find deterministic correspondences between samples from different domains (e.g., source and target languages)
- Manifold learning & geometry processing
 - Use transport maps to compute geodesics, morph shapes, or align 3D meshes
- Generative modeling
 - Map latent Gaussian samples to complex data distributions (e.g., in normalizing flows, transport-based samplers)
- Biological & medical data analysis
 - Align spatial transcriptomics, single-cell datasets, or patient cohorts under interpretable transport maps
- Gradient flows and PDEs
 - Solve evolution equations (e.g., Fokker–Planck) via dynamic OT paths

Interactive Demo: Gaussian Monge

Implement closed-form solution Monge map for 2D Gaussians



$$\alpha = \mathcal{N}(\mu_\alpha, \Sigma_\alpha)$$

$$\beta = \mathcal{N}(\mu_\beta, \Sigma_\beta)$$

$$A = \Sigma_\alpha^{-\frac{1}{2}} (\Sigma_\alpha^{\frac{1}{2}} \Sigma_\beta \Sigma_\alpha^{\frac{1}{2}})^{\frac{1}{2}} \Sigma_\alpha^{-\frac{1}{2}}$$

$$T(x) = \mu_\beta + A(x - \mu_\alpha)$$

